

Experiment 20: Image Sharpening using Laplacian Mask (Negative Center Coefficient)

AIM

To sharpen an image using a Laplacian mask with a negative center coefficient.

PROCEDURE

1. Read the image.
2. Create a Laplacian kernel.
3. Apply the filter using filter2D().
4. Display the sharpened image.

PROGRAM

```
import cv2

import numpy as np

img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

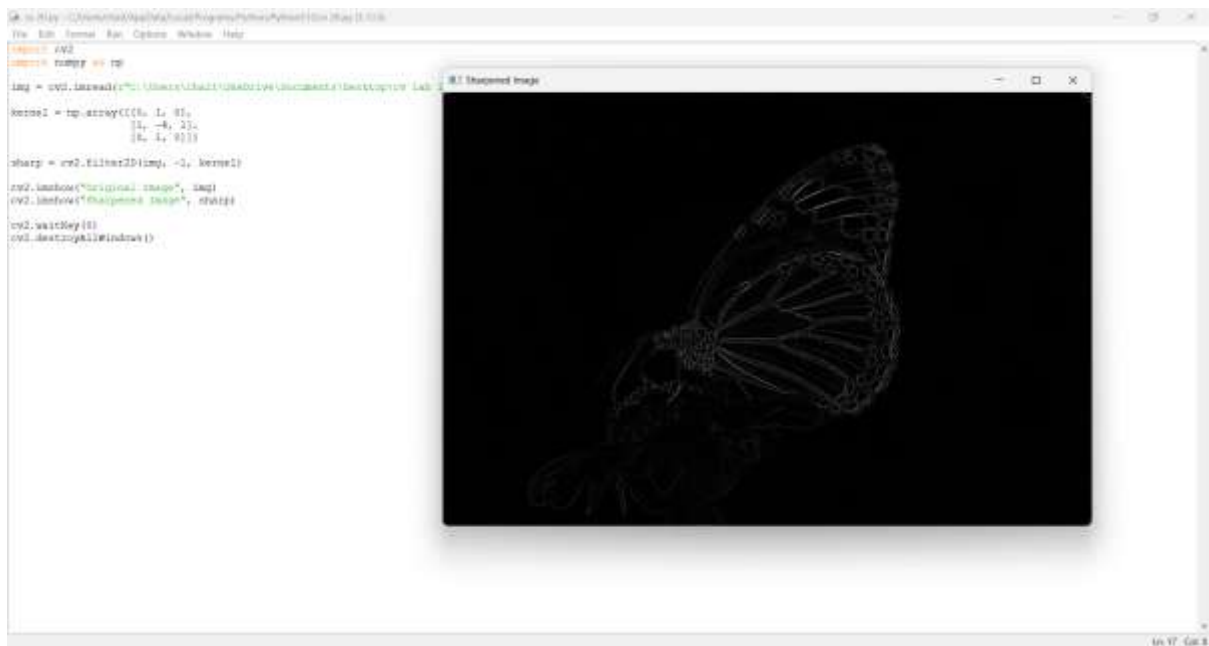
kernel = np.array([[0, 1, 0],
                   [1, -4, 1],
                   [0, 1, 0]])

sharp = cv2.filter2D(img, -1, kernel)

cv2.imshow("Original Image", img)
cv2.imshow("Sharpened Image", sharp)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT



Explanation

We applied a Laplacian mask with a negative center coefficient to enhance the image details. This sharpening technique makes object edges and fine features more prominent.